

Question 1-Process

Write a C program that creates a child process using fork(). Print:

- Parent PID
- Child PID
- Parent Process ID (PPID) of child

Question 2-Process

Write a program where:

- parent prints numbers 1–5
- child prints numbers 6–10

Observe execution order.

Question 3-Process

Create two child processes from one parent process and display the process tree.

Question 4 — FCFS Scheduling

Write a program to implement **First Come First Serve (FCFS)** CPU scheduling algorithm and calculate:

- Waiting Time
- Turnaround Time
- Average Waiting Time
- Average Turnaround Time

Question 5 — SJF Non-Preemptive

Implement **Shortest Job First (Non-Preemptive)** scheduling algorithm.

Display:

- Gantt Chart
- Waiting Time
- Turnaround Time

Question 6 — SJF Preemptive (SRTF)

Write a program for **Shortest Remaining Time First (SRTF)** scheduling.

Calculate:

- Completion Time
- Waiting Time
- Turnaround Time

Question 7 — Round Robin Scheduling

Implement Round Robin Scheduling using a user-defined time quantum.

Display:

- Gantt Chart
- Waiting Time
- Turnaround Time

Question 8 — Priority Scheduling

Write a program to implement:

- Non-preemptive priority scheduling
- or
- Preemptive priority scheduling

Calculate:

- Completion Time
- Waiting Time
- Turnaround Time

Question 9 — Banker's Algorithm

Write a program to implement the Banker's Algorithm for deadlock avoidance.

The program should:

- take Allocation, Maximum, and Available matrices as input
- determine whether the system is in a safe state
- display the safe sequence

Question 10 — Deadlock Detection

Write a program to detect deadlock in a system using:

- Allocation matrix
- Request matrix
- Available resources

Question 11 — FIFO Page Replacement

Write a program to implement the First-In First-Out (FIFO) page replacement algorithm.

The program should:

- accept page reference string
- accept number of frames
- count page faults and page hits

Question 12 — LRU Page Replacement

Implement the **Least Recently Used (LRU)** page replacement algorithm.

Display:

- frame status after each page reference
- total page faults